

Dingning Cao

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Education

Massachusetts Institute of Technology

Sept 2024 – May 2028

BS in Computer Science + Arts and Design (GPA 4.6)

- Designer of MIT 2028 Brass Rat; MITIT, Reality Hack, Infinite Magazine Art Director, HackMIT Marketing
- **Relevant Coursework:** Game Design, Computer Graphics, Algorithms, Linear Algebra, Machine Learning

Milton Academy

Sept 2020 – June 2024

Skills

Programming: Python, C# (Unity), JavaScript, HTML/CSS, React, Java, PyTorch, scikit-learn, ComfyUI, C++

Design & Modeling: Figma (UI/UX), Adobe Suite, Procreate, Blender (3D Modeling), Maya, Rhino

Research | Cambridge, MA

MIT CSAIL Researcher (with Antonio Torralba & Yael Vinker)

Feb 2025 - Present

- Co-first author of *Doodle Agent: Exploring Freeform Visual Generation with Multimodal LLMs* (ICCV AI4VA Accepted), built multimodal agent framework for interactive drawing with human evaluation of style, structure, and expressiveness.
- Developed full-stack [SketchAgent](#) platform (Flask, React, SVG), enabling sequential prompting, live visualization, and reproducible ML experiments.

MIT HCIE Group Researcher (with Jiaji Li & Stefanie Mueller)

May 2025 - Present

- Created four game applications in Unity with Blender-modeled assets, showcasing rigid-soft hybrid bending structures with embedded sensing.
- Engineered sensor data processing and interactive visualization, developing modular prototypes for evaluating tangible input methods.
- Publications: *Y-zipper: 3D Printing Flexible-Rigid Transition Mechanism for Rapid Assembly* (CHI 2025 Accepted); *Xspine: Integrating Motion Sensing Capability into Dynamic Structures Using Multi-material FDM 3D Printing* (UIST 2025 Posters accepted, CHI 2025 Accepted).

Projects & Experience

AI Researcher - Lica World

2026-Present

- Fine-tune models for layout generation for poster, presentation design.
- Research benchmarks in slide deck generation (combining solo Pillar with overall cohesiveness).

AI Technical Artist - Mihoyo (Genshin)

2025-2026

- Trained Mihoyo-specific LoRas and built pipelines via ComfyUI
- Created an independent Agent-powered tool that aids upstream design for Impact.

Game Development Projects

2023-2025

- [Threadspace](#) (Godot, GMTK 2025) Designed mechanics, art assets, and 8 levels with OOP components; coordinated playtests. ★ Ranked #48/9600 (Top 0.5%), featured at Boston Indie Game Festival.
- [Pizzapocalypse!](#) (Unity, iOS) Implemented tilemap levels, NPC AI, UI systems, and progression mechanics. ★ Published on iOS (4.8 rating); won NHSGA Mass Appeal Prize.
- [Tea Time in Five](#) (Unity, Brackeys Jam 2025, 3D) Modeled Blender art assets and coded movement systems; managed team and playtesting. ★ Ranked 236/2299.

Creative Web Applications

2024-2025

- **Dreamscape:** Full-stack journaling platform (React, Node.js, MongoDB) with media storage and collaborative editing. ★ Placed 5th in MIT WebLab.
- **Mr. Li-Po:** Sketch-to-text pipeline with HTML5 Canvas, Node.js, and API-driven generative text.
- **Ablecademy:** iOS+iPad app built in Swift to assist children with Down syndrome; conducted user testing.

Awards

- USA Computing Olympiad Gold Division — Jan 2024
- VEX Robotics Regional Skills Champion & Design Award (World #12) — Feb 2024